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Heatmaps 2

Heatmaps

This Help Center article explains how to generate static and dynamic Heatmaps.

When to use Heatmaps

Heatmaps allow visualization of aggregated eye tracking data, *which indicates where respondents looked within the visual stimulus.*

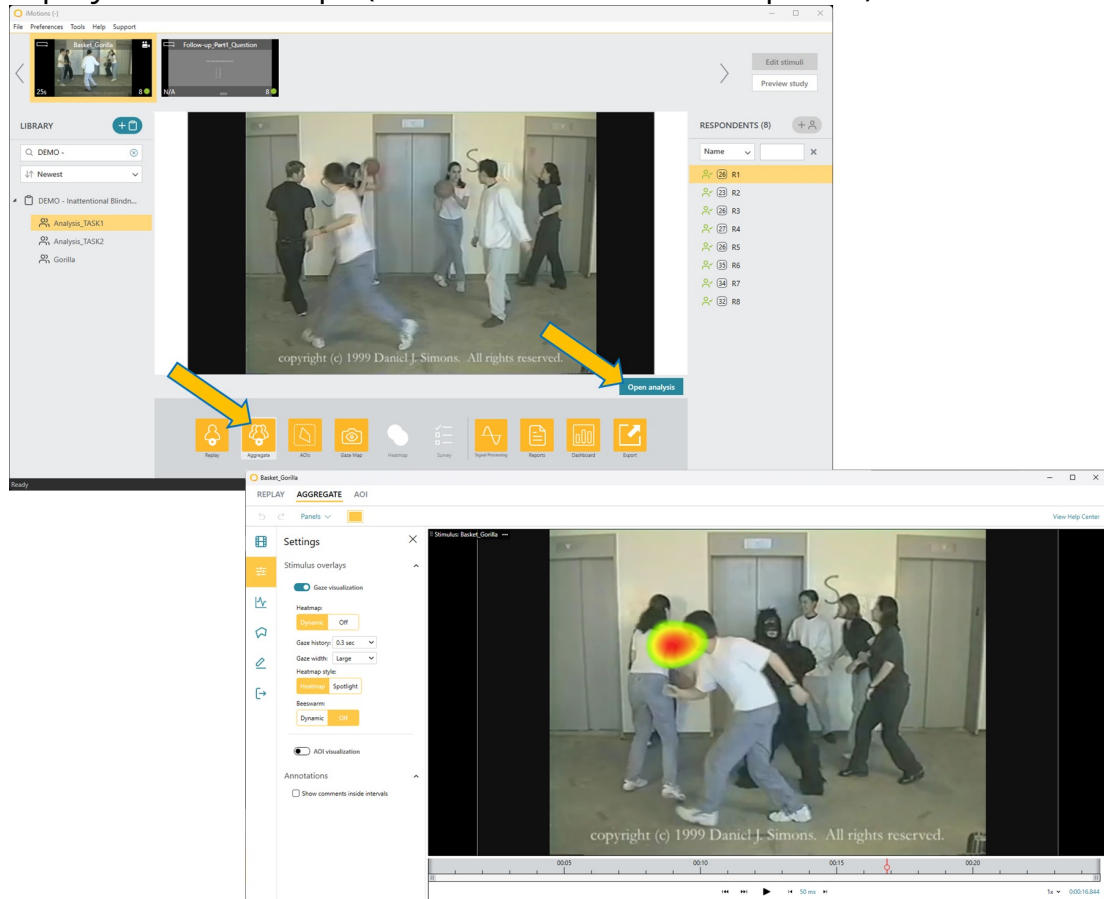
- Heatmaps can be displayed on static images (summarizing the aggregate eye tracking data in an image) or in replay videos (with aggregate eye tracking data overlaid).
- Heatmaps allow qualitative assessments, such as identifying the products on a store shelf or parts of a website that received the most attention or were overlooked.
- Note: Heatmaps are only available for 'static stimuli' that are the same for all respondents, where data can be aggregated (as opposed to 'dynamic stimuli', e.g. screen recordings or eye tracking glasses)

How to Create and View Heatmaps

Heatmaps are available in any iMotions analysis that includes aggregated eye tracking data. Static Heatmap can be found in the main iMotions window; Static and Dynamic Heatmaps can be found in the Aggregate Replay window.

1. **Select study** with eye tracking data.
2. **Select or create analysis.**
3. Select a stimulus with aggregated eye tracking data.
4. Click Heatmap to display the resulting heatmap.
 - a. This is only available for static stimulus types (e.g., images, survey slides)
 - b. For dynamic stimuli, static stimuli and Heatmaps can be created with [Gaze Mapping](#) or [Image Scenes](#).

5. **Select Aggregate** and click **Open Analysis** to open the aggregate replay with Heatmaps (and other visualization options).



Heatmaps in Aggregate Replay

Heatmaps can be visualized and exported from within the [Aggregate Replay window](#). Other stimuli can be selected from the Aggregate window but will only display static stimuli.

Visualization Settings menu

The **Settings menu** (on the left of the Aggregate Replay window) contains visualization settings for eye tracking data, including Heatmaps.

- **Stimulus overlays** menu provides settings for gaze visualization
- **Gaze history** settings can be used to select the time window for visualized data (relative to the selected time in the replay)
- The **Export menu** can then be used to create export files according to the visualization settings (i.e. Heatmaps):
 - The Video and Image options create images (.png) or videos (.wmv, .mp4), as they appear in the Aggregate Replay window.

- Additional information about Video and Image Exports from Replays



Heatmap Options

Static Heatmaps

Static heatmaps visualize the aggregate eye tracking data *as an image*, summarizing the eye tracking data of all respondents included in the analysis across the entire duration of the stimulus. The parts of the stimulus that were looked at most by the respondents, color red. The parts of the stimulus that were looked at least by the respondents, remain transparent. The colors green, yellow and orange represent everything in between those extremes.

Thereby, a combination of two factors contributes to the coloring of the heatmap: **(1)** *how many* respondents looked at something, and *how long* each of the individual respondents looked at it.

These heatmaps can only be generated for static stimulus types (e.g., images, survey slides, 'gaze-mapped images', or 'image scenes'). Therefore, in order to generate static heatmaps for *dynamic* stimulus types (e.g., videos, websites, screen recordings, etc.), you should first apply Gaze mapping (see also: [Gaze mapping](#)) or use the Scene editor to cut 'image scenes' (see also: [Scene editor](#)).

Dynamic Heatmaps

In contrast to static heatmaps, dynamic heatmaps are available for both static and dynamic stimulus types. In other words, for dynamic heatmaps it is not necessary to first apply Gaze mapping, or to cut 'image scenes'.

Dynamic heatmaps are similar to static heatmaps, but rather than to summarize the aggregate eye tracking data *over time* (as one image), they represent the aggregate eye-tracking data *at every point in time* (as a video).

That means that the interpretation of dynamic heatmaps is slightly different from the interpretation of static heatmaps, as, at any point in time, the coloring of the dynamic heatmaps is only determined by *how many* respondents looked at something; not *how long* the individual respondents looked at it.

Dynamic heatmaps are not so much about summarizing *which parts of a stimulus* were looked at most, but rather about understanding *how consistent the gaze behavior is* within a certain group of respondents while interacting with the stimulus. For instance, expert readers of sheet music would exert a much more consistent reading pattern while listening to a piece of music than novices would do; for novices the reading of the notes is much more challenging, and therefore, their eye tracking behavior would be much less consistent.

Customization options

Heatmap style	Choose between a heatmap and spotlight style
Beeswarm	Choose to add a beeswarm (representing <i>all</i> fixations or <i>all</i> respondents)

These options can freely be combined, such that it is possible to generate a *static* heatmap with *dynamic* beeswarm, or vice versa, a *dynamic* heatmap with *static* beeswarm.

Heatmaps for individual respondents

Heatmaps are generated at the *aggregate* level and always represent *all respondent included in the analysis*. That means that it is only possible to generate heatmaps from individual respondents using a work-around. Specifically, you can generate heatmaps for individual respondents by creating analyses that include only one respondent.

Note that, instead of generating heatmaps for individual respondents, you may also consider generating scan-paths for individual respondents (see also: **Scan-paths**). In fact, scan-paths would often be more informative than heatmaps.

